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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEM

WINTER SEMESTER - 2024

Digital Assignment – 1

Course Name : Information System Management

Course Code : CSE3502

Slot : L51+L52

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1. State the differences between static and dynamic IP addresses.

- Command for setting the IP address of a client.
- Command to view the subnet details.
- Command for complete details of the network adapter.
- Renewal of IP addresses after the timestamp has expired.

Difference between Static and Dynamic IP addresses

S.NO	Static IP Address	Dynamic IP address
1.	It is provided by ISP(Internet Service Provider).	While it is provided by DHCP (Dynamic Host Configuration Protocol).
2.	Static ip address does not change any time, it means if a static ip address is provided then it can't be changed or modified.	While dynamic ip address change any time.
3.	Static ip address is less secure.	While in dynamic ip address, there is low amount of risk than static ip address's risk.
4.	Static ip address is difficult to designate.	While dynamic ip address is easy to designate.
5.	The device designed by static ip address can be traced.	But the device designed by dynamic ip address can't be traced.
6.	Static ip address is more stable than dynamic ip address.	While dynamic ip address is less stable than static ip address.
7.	The cost to maintain the static ip address is higher than dynamic ip address.	While the maintaining cost of dynamic ip address is less than static ip address.
8.	It is used where computational data is less confidential.	While it is used where data is more confidential and needs more security.
9.	Simplifies the troubleshooting as the ip is always the same.	While dynamic ip increases the complexity of diagnosing the network issues.

Command for setting the IP address of a client.

To set the IP address for the client if he is connected to DHCP server:

netsh interface ipv4 set address "InterfaceName" dhcp – this command obtains its IP address dynamically from a DHCP server.

To set the IP address for the client if he is connected to static server:

netsh interface ipv4 set address "InterfaceName" static IPAddress SubnetMask DefaultGateway

Command to view the subnet details.

"ipconfig /all" – command can be used to view the subnet mask details.

```
C:\Users\Administrator>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : SJT806SCORE070
Primary Dns Suffix . . . . . : VITUNIVERSITY.LOCAL
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : VITUNIVERSITY.LOCAL
```

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (17) I219-LM
Physical Address. . . . . : 38-CA-84-44-05-B6
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6535:1937:59e4:99d5%13(Preferred)
IPv4 Address. . . . . : 10.30.146.80(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.30.146.1
DHCPv6 IAID . . . . . : 104385156
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-27-9C-42-38-CA-84-44-05-B6
DNS Servers . . . . . : 10.10.4.10
                        10.30.2.213
NetBIOS over Tcpip. . . . . : Enabled
```

Command for complete details of the network adapter.

“**ipconfig /all**” – command can be used to view all the details of the network adapter.

```
C:\Users\Administrator>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : SJT806SCORE070
Primary Dns Suffix . . . . . : VITUNIVERSITY.LOCAL
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : VITUNIVERSITY.LOCAL
```

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Ethernet Connection (17) I219-LM
Physical Address. . . . . : 38-CA-84-44-05-B6
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6535:1937:59e4:99d5%13(Preferred)
IPv4 Address. . . . . : 10.30.146.80(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.30.146.1
DHCPv6 IAID . . . . . : 104385156
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-27-9C-42-38-CA-84-44-05-B6
DNS Servers . . . . . : 10.10.4.10
                        10.30.2.213
NetBIOS over Tcpip. . . . . : Enabled
```

Renewal of IP addresses after the timestamp has expired.

“**ipconfig /renew**” – command can be used to renew the IP address.

2. There is a trouble ticket raised by your clients that the intranet is not working. You know that the local network is a Gigabit Ethernet. How would analyse this Ethernet status and analyse all the protocols used?

Solution:

To troubleshoot the reported issue with intranet not working on a Gigabit Ethernet we can follow these steps:

1. With the help of WEB TERMINAL python runtime application, if client have already pre-installed the application in his system and tell him to share the web link of the terminal.
2. Check Physical Connections: Ensure all ethernet cables are securely connected and verify that the network devices are powered on and functioning correctly.
3. Check the status of the network adapter by using the "**ipconfig /all**" command to verify if the network adapter is properly configured with an IP address, subnet mask and default gateway.
4. Perform ping test using the "**ping -n 1 -l 1 IPaddress**" which is an optimal way of testing the network and ensure the connectivity.
5. Ensure the client has obtained a valid IP address lease from the DHCP server by using the "**ipconfig /all**" command which displays the lease obtained and lease expire date along with other information.
6. Using the "**netstat -e**" command which displays the TCP/IP network protocol Statistics and information. It shows the number of packets which are sent and received. If there are no packets sent or received, we can conclude that the internet is not working.
7. Using the "**tracert -d target_name**" command can be used to identify the potential issues along the route.
8. Using the "**nslookup**" command to troubleshoot the DNS related problems if it exists.
9. Using the "**net config**" command to view information about the configuration of the client's system.